BridgeFinder: Finding Research Collaborators Across Communities at UIUC

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Abstract

Students who want research experience often know an instructor but few researchers outside that instructor's circle. BridgeFinder helps them find related researchers and see who might connect them. We have built a three-screen prototype, analyzed two UIUC publication samples, and gathered feedback from five students. Our historical test finds that shared-contact ranking predicts more later collaborations than our method. The next study will test whether students nevertheless find our suggestions relevant, unfamiliar, and useful for contacting someone.

1 Objective

A student who enjoyed a computing class might want to apply those skills in biology or agriculture. Searching for papers can reveal relevant names, but it does not tell the student whom they could approach or how to start a conversation.

BridgeFinder helps UIUC undergraduate and early-stage graduate students explore research beyond their existing contacts. A student chooses a familiar researcher, sees suggestions outside that person's collaboration group, and opens a card with topics, papers, and a possible introduction route. The starting researcher's work stands in for the student's interests, so the student needs no publications of their own.

The challenge is to suggest someone relevant who also extends the student's usual circle. The prototype implements this flow; whether it helps students choose useful contacts still needs testing.

2 Related Work

Researcher search. Illinois Experts [12] collects institutional profiles and publications, making expertise easy to check. Google Scholar [9] helps readers find papers and follow authors. Both help identify names; our task also needs a route from a familiar person to a suggested contact. BridgeFinder puts that relationship beside the research match.

Literature maps. Connected Papers [10] makes related papers visible in a local map. ResearchRabbit [11] supports continued discovery from paper collections. Both make exploration easier, but their main objects are papers. BridgeFinder adapts visual exploration to people and possible introductions.

Recommendations. Liben-Nowell and Kleinberg [5] compare network-based link predictions. Shared-contact measures are understandable, but favor nearby people. We compare against Adamic–Adar while looking farther away. Tang et al. [8] combine research interests and relationships for cross-domain collaboration recommendation. That supports considering both topic and network evidence, but ranking quality alone does not show whether a student can use a suggestion. We therefore also plan a student evaluation.

Network theory. Granovetter [4] explains how weak ties can reach beyond close circles. Burt [2] explains the advantage of bridging separate groups. These ideas motivate our discovery goal; they do not establish that a co-authorship path is a trusted introduction.

3 Network Analysis

3.1 Graph and communities

We use OpenAlex [7] publications from 2015–2024. Nodes are authors affiliated with UIUC on a collected paper. An edge joins two authors who wrote together, weighted by their number of joint papers. We retain papers with at least two UIUC authors and omit external collaborators.

Our pilot contains 5,000 collected works, of which 2,493 meet that rule. The larger sample contains 25,000 works, of which 13,229 remain. These are samples, not a complete campus directory. Table 1 describes the earlier papers used to make recommendations.

We use Louvain [1] to find communities with many collaborations inside each group. It seeks high *modularity*: more within-group edge weight than expected from node strengths. We chose it over Girvan–Newman [3], whose repeated shortest-path calculations are costly on these graphs.

The high modularity suggests concentrated collaboration groups, making cross-group discovery worth exploring.

Table 1: Co-authorship graphs from 2015–2019 papers.

	Pilot	Larger
Authors	3,642	9,042
Edges	16,408	36,748
Communities	344	455
Modularity	0.900	0.905

Disconnected components also raise modularity, so this is not evidence of disciplinary separation. Communities are numbered groups, not verified departments.

3.2 Choosing suggestions

Following the proposal, candidates must be in another community and at least three hops from the starting researcher. One hop is one co-authorship edge. This excludes existing collaborators and people who already share a co-author. Candidates must also be reachable so we can show a route. All 500 suggestions in the current 50-person demo have three-edge routes.

We rank candidates by topic overlap, with a bonus for introduction paths:

$$\text{Score}(u, v) = S(u, v) + 0.5\frac{c_3(u, v)}{1 + c_3(u, v)}. \quad (1)$$

S is the number of shared topics divided by the number of distinct topics across both researchers. Each researcher has up to ten topics from their 2015–2019 papers. c_3 counts three-edge paths through two distinct intermediaries. The bonus grows with the number of routes but stays below 0.5. More distant candidates remain eligible without that bonus. The coefficient and saturation are fixed implementation choices, not learned probabilities.

Without the network, the app would lose its group boundary, distance rule, and introduction routes and become a topic search tool. These rules help us look beyond shared contacts; they do not measure personal trust.

4 Design and Implementation

The working prototype has three views:

1. Researchers. The student searches 50 starting researchers by name or topic. We chose highly connected authors across large communities so the demo has enough suggestions. As in profile search [12], this provides a familiar entry point without requiring a student profile.

2. Network. A map shows the selected researcher, suggestions, and intermediaries. Solid lines show observed co-authorships; orange dashed lines show suggested connections. This borrows visual exploration from literature maps [10, 11] while focusing on people.

3. Suggestions. Selecting a name or dashed line opens a card with an observed shortest route, topics, and example papers. The route appears first in response to student feedback. An optional explanation shows the ranking score. An outreach draft can be edited, copied, or downloaded; no email is sent. Figure 1 shows the map and route detail.

The app, historical test, and study cards share one ranking implementation. Recommendations are computed in Python and displayed in D3. The core flow works on desktop and mobile, including keyboard navigation, and has been browser-tested for all 50 starting researchers.

5 Users and Feedback

Our target users want research opportunities but have few contacts beyond their classes or department. We gathered feedback in five individual, 30-minute think-aloud sessions: three CS/CE undergraduate juniors or seniors and two graduate students from Bioengineering and Information Sciences.

The undergraduates fit the entry problem of exploring research before having publications. The graduate students add perspectives from application areas. Five sessions suit an initial search for interaction problems [6], rather than a campus-wide usability estimate. The feedback led to three changes:

Getting started. Students without publications could not use themselves as the starting node. We added a searchable researcher list so a familiar instructor could serve as the entry point.

Reading the map. Students interpreted dashed lines as weak existing collaborations. We added a legend that separates observed co-authorships from suggestions, as shown in Figure 1.

Choosing a next step. Students looked for who might introduce them before reading papers and topics. We put the route first and moved technical scores out of the main card. The current ranking explanation is optional.

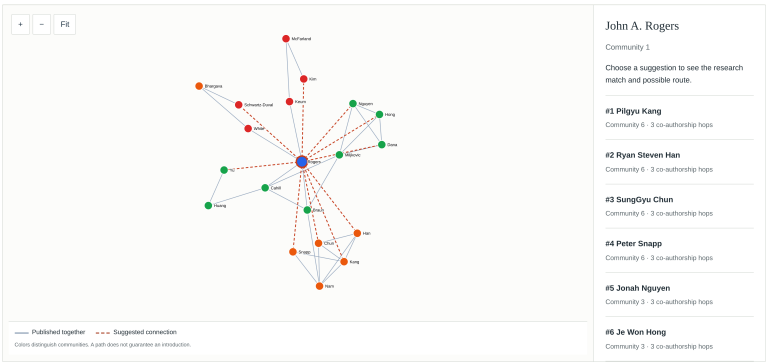
These qualitative observations guided the design. They do not show that the revisions solved the problems; the next round will check that directly. We do not report issue counts or participant quotations.

6 Evaluation

6.1 Historical prediction

We used 2015–2019 papers to build the graph and topic profiles, then checked whether suggestions became new cross-community co-authors in 2020–2024. We evaluated 184 starting researchers in the pilot sample and 1,369 in the larger sample. Each had at least one new cross-community tie between training-graph authors. Researchers without such future ties are excluded.

We compare with **Topic-Only** on the same candidate pool and **Adamic–Adar**, which ranks shared contacts at distance two. Restricting Adamic–Adar to our three-or-more-hop pool would give only zero scores. All methods predict the same future ties, so this compares discovery strategies with different pools.



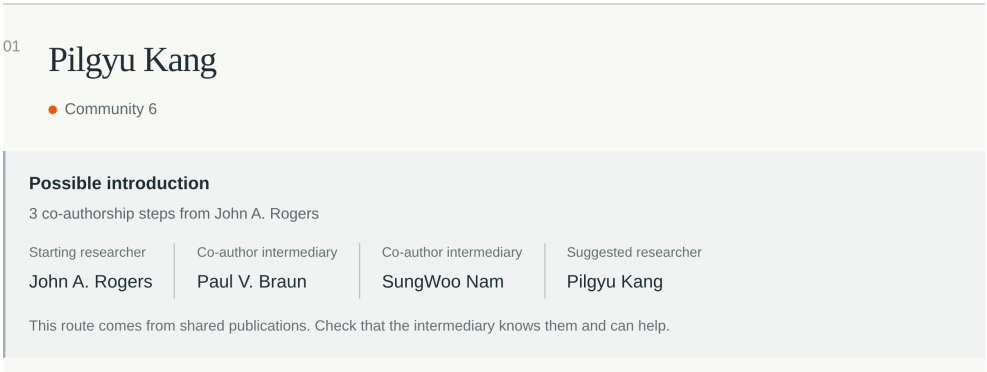


Figure 1: Top: John A. Rogers's map, with observed co-authorships in solid lines and suggestions in orange dashed lines. Bottom: Pilgyu Kang's card detail shows a route through Paul V. Braun and SungWoo Nam. Topics and papers appear below this detail. The route is observed in publications, not a confirmed introduction.

P@5 is the fraction of the first five suggestions that became collaborators. **NDCG@10** rewards placing actual collaborators near the top of the first ten. Both are averaged across starting researchers. These measures test shortlists and ordering, respectively.

Table 2: Prediction of new 2020–2024 cross-community ties. Higher is better.

	Pilot		Larger	
Method	P@5	NDCG @10	P@5	NDCG @10
Topic-Only	0.0043	0.0154	0.0108	0.0205
Adamic–Adar	0.0130	0.0368	0.0282	0.0472
BridgeFinder	0.0065	0.0149	0.0102	0.0225

Table 2 shows that BridgeFinder predicts fewer later ties than Adamic–Adar. It misses our targets of doubling P@5 and adding 0.05 to NDCG@10. The path bonus also has mixed results against Topic-Only. We have not estimated uncertainty, and these comparisons do not isolate distance from scoring. Historical prediction does not establish whether students find the farther-away contacts useful.

6.2 Planned student study

We will recruit 15 students, aiming for three each from CS, ECE, Bioengineering, Crop Sciences, and Information Sciences. This broadens the formative sample but remains exploratory.

Each student will choose a starting researcher, open a suggestion and outreach draft, and explain the route and dashed line without coaching. A facilitator will record task completion and whether the student identifies the intermediaries and distinguishes suggestions from existing ties.

Students will then choose one of five study contexts and rate up to 15 shuffled cards: five from each method above. Method labels are hidden. Shared candidates appear once and count for every method that selected them. We will measure:

- **Relevant discovery:** ratings from 1 to 7 for relevance to the starting researcher's work and novelty to the student. We target means of at least 5 and a 25-percentage-point gain over Adamic–Adar in cards rated at least 5 on both. Topic-Only is a second comparison.
- **Willingness to contact:** a 1–7 rating of how realistic contact feels given the route. We target a mean of at least 5 and improvement over Topic-Only. This tests the full suggestion, not the path alone.

- **Usability:** at least 12 of 15 students should complete the task and explain the line types correctly without help. The ten-item System Usability Scale will assess the interactive app, with a target mean of 80/100.

We will compare methods using participant averages, not treat individual cards as independent responses. The rating and analysis tools are implemented; the study has not yet been run.

7 Limitations and Next Steps

Co-authorship cannot confirm an introduction or mentoring availability. Our sample omits external and informal ties and falls short of the proposed 110,000-work collection. The 50 starting researchers and five study contexts limit coverage. A researcher's topics may also be a poor proxy for a student's interests.

In November, we will run the student study and revise the interface. Any ranking changes will use separate validation data. The final question is whether students can find a useful contact beyond their usual circle.

Contributions and AI Use

Evan Lin worked on data collection, prediction tests, and the map. Eric Liu worked on scoring, citation analysis, study tools, and formative sessions. Both contributed to the report. Claude assisted with early development and drafting; OpenAI Codex assisted with checks, tests, and revision. No participant responses or quotations were generated.

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